

Tilings

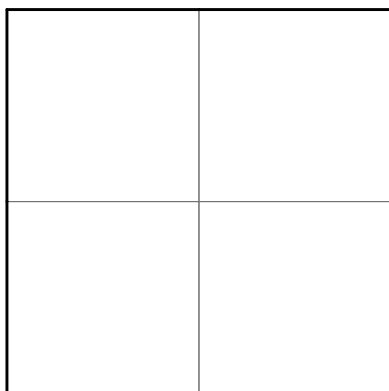
Week 1 · Grades 2–3

A **tiling** covers a shape completely: no gaps, no overlaps, nothing sticking out past the edge. Today's piece is the **domino**. It covers two squares that share a side, and you may turn it any way you like.

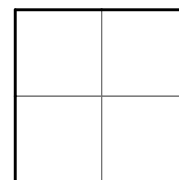
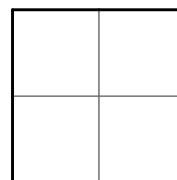
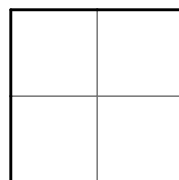
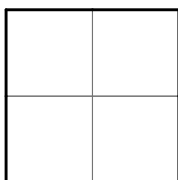


1. How many ways?

Cover the 2×2 board with dominoes. Now find a *different* way. Draw every different way you find in the small boards.



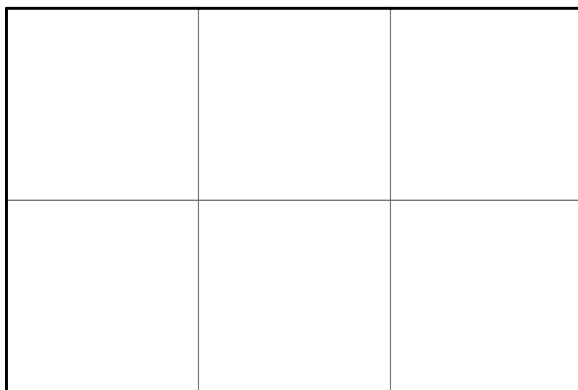
1 inch



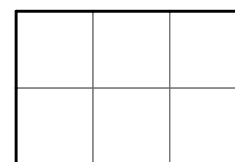
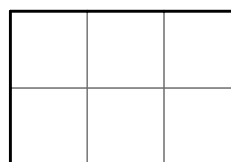
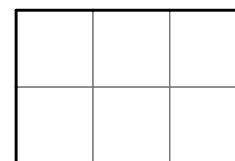
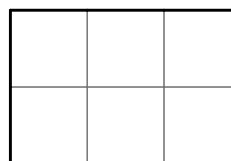
draw here

Number of different ways for the 2×2 board: _____

Now the 2×3 board. Draw every different way.



1 inch



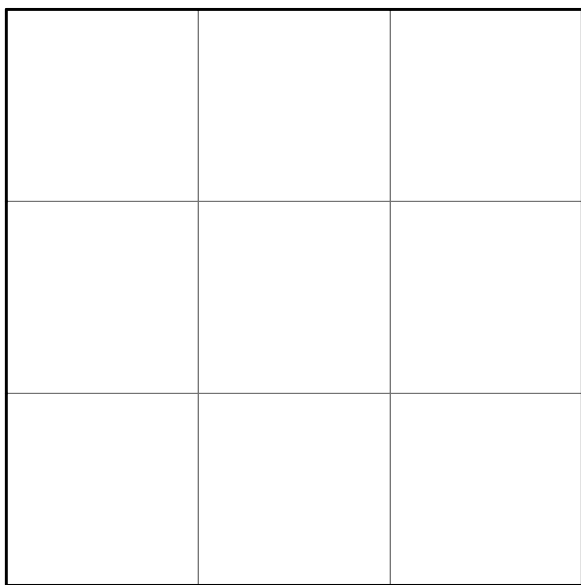
draw here

Number of different ways for the 2×3 board: _____

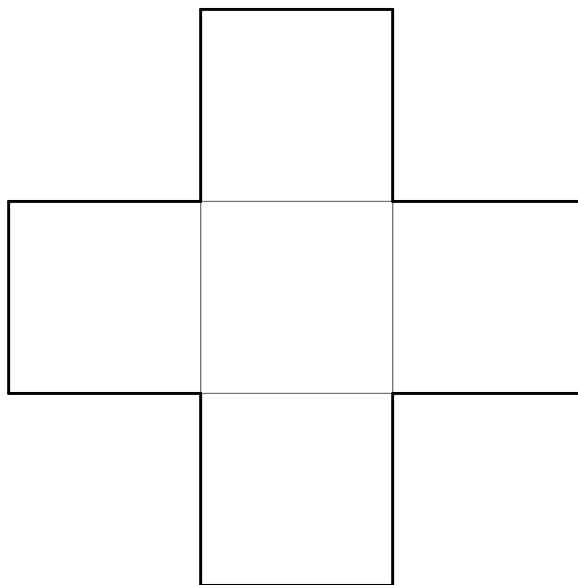
If you turn a covered board around, or flip it over, does it count as the *same* way or a *different* way? Decide with your table and stick to it: ☐ same ☐ different

2. Can it be covered?

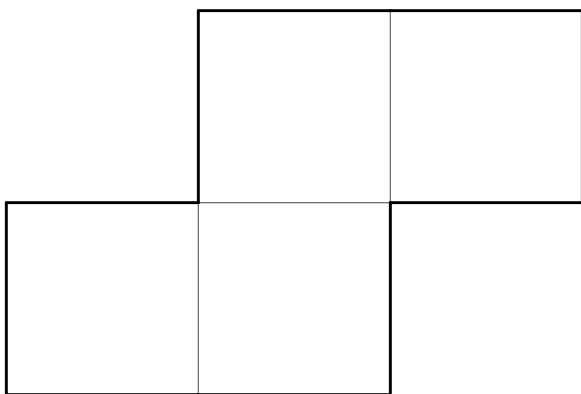
Try to cover each shape with dominoes. If it works, draw the dominoes. If you are *sure* it cannot be done, write NO under the shape and be ready to say why you are sure.



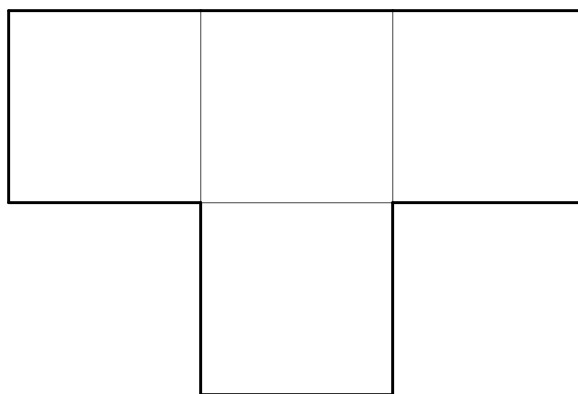
A



B



C



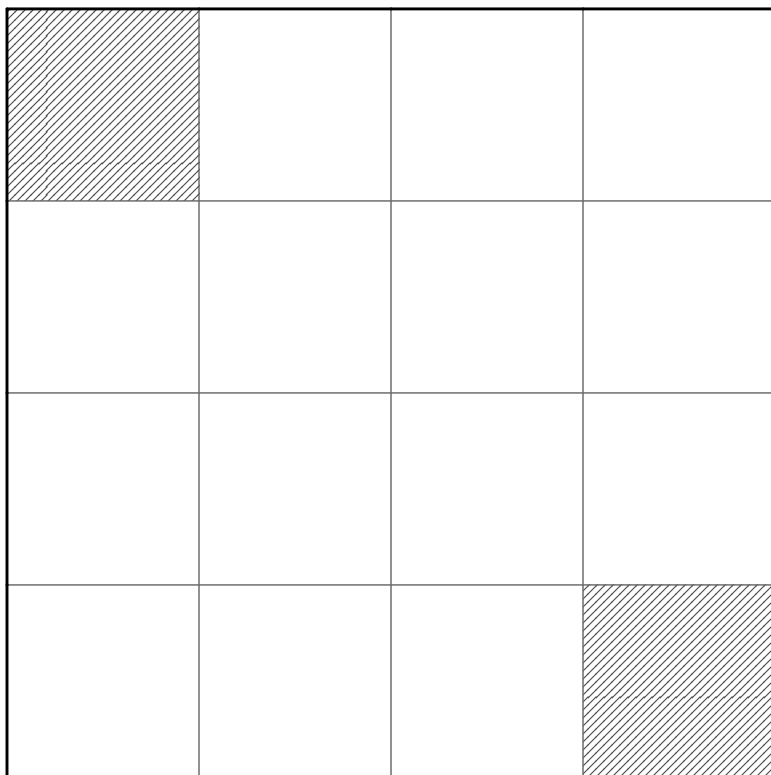
D

shape	how many squares?	can it be covered?
A		
B		
C		
D		

Rule 1. A shape with an _____ number of squares can never be covered with dominoes, because

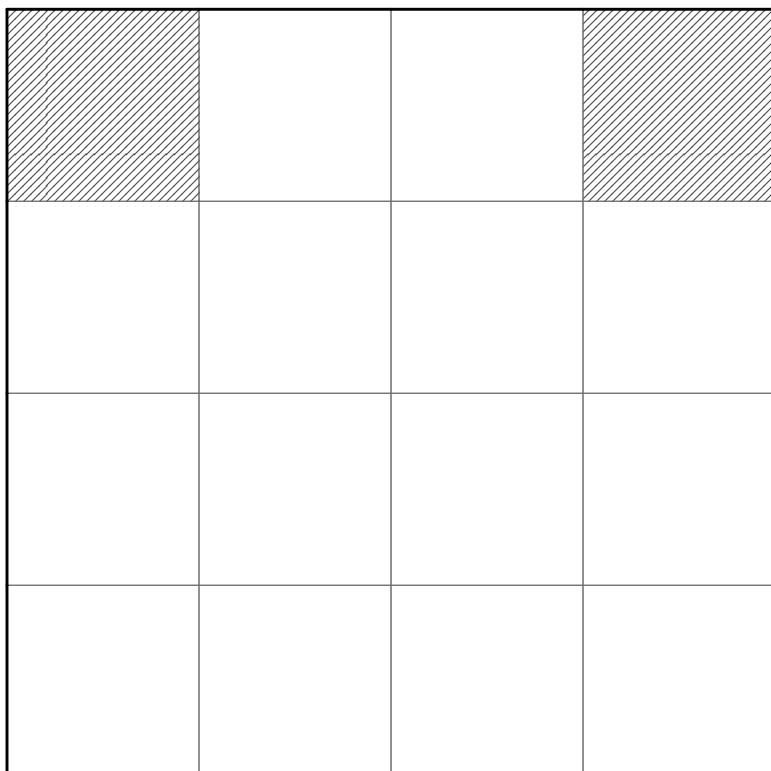
3. Two boards with 14 squares

Both boards lost two corners (the striped squares are gone), so each has 14 squares. That is an even number, so Rule 1 does not stop us. Try both. (Boards are at 1-inch scale: use the paper dominoes.)



Board E

☐ yes ☐ no



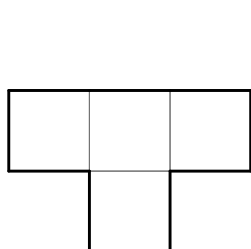
Board F

☐ yes ☐ no

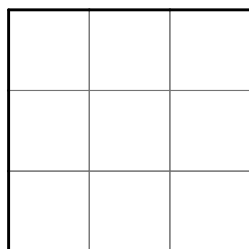
One of these boards can be covered and one cannot. But how could you be *sure* that it cannot, without trying every possible way? Turn the page when your table has an idea, or is stuck.

4. The secret coloring

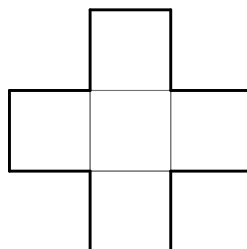
Color each shape like a checkerboard: squares that share a side get different colors. (Start with any square, shade it, then shade every other square.) Then count. The striped squares are the missing corners: they are not part of the board.



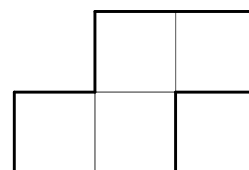
D



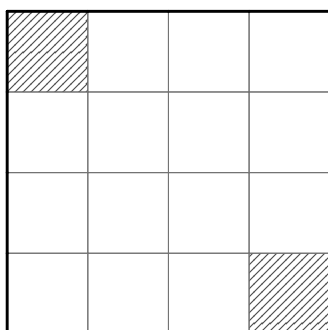
A



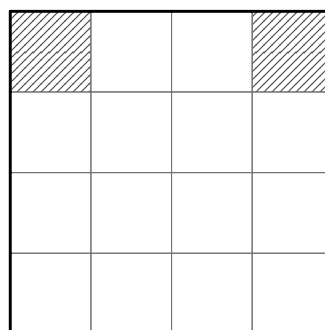
B



C



E



F

shape	shaded squares	white squares	can it be covered?
D			
A			
B			
C			
E			
F			

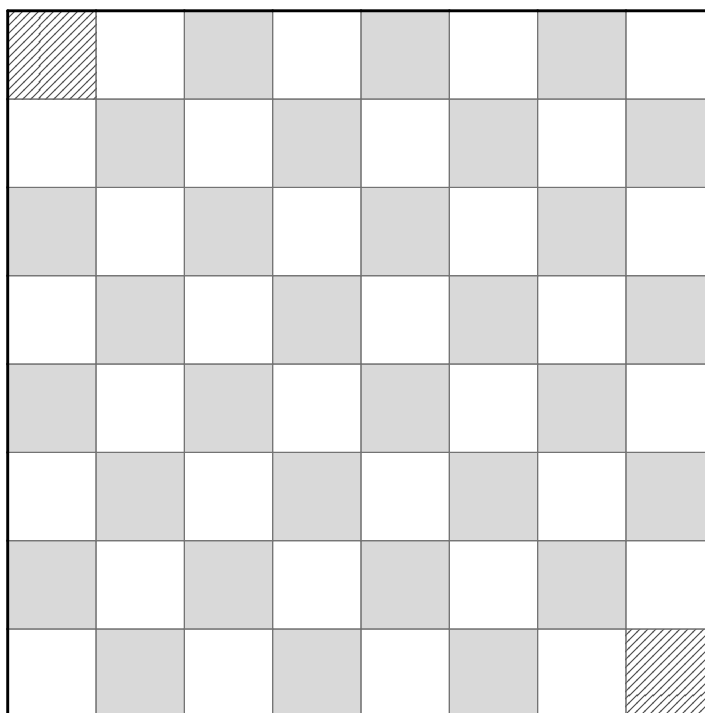
Put a domino anywhere on a checkerboard. It always covers _____ shaded square(s) and _____ white square(s). So 7 dominoes always cover _____ shaded and _____ white squares.

Rule 2. A shape can only be covered by dominoes if

Explain in one sentence why shape D cannot be covered, even though it has 4 squares. Then do the same for Board E.

5. The big board

This chessboard lost two opposite corners. It has 62 squares, so 31 dominoes would be needed. Count the shaded squares and the white squares. (Find a fast way. Do not count them one by one.)

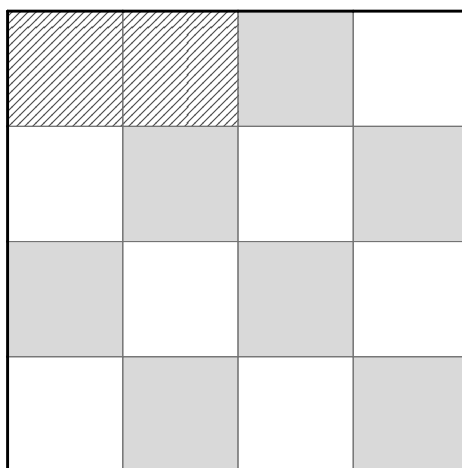


shaded: _____ white: _____ Can it be covered? ☐ yes ☐ no

Write a reason that would convince a friend who wants to keep trying:

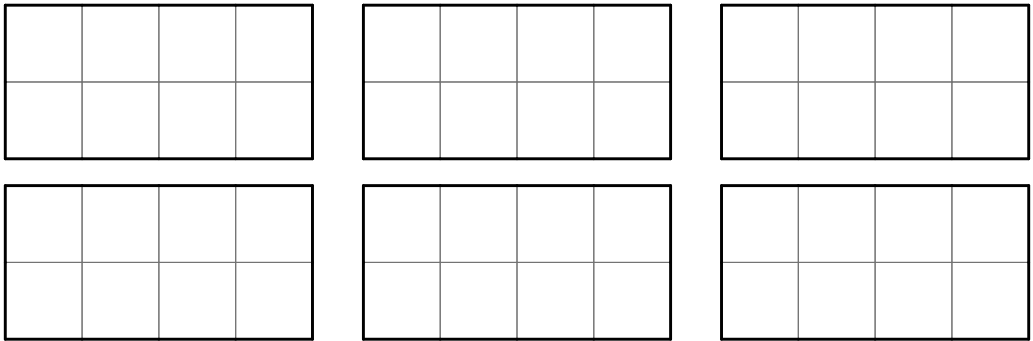
6. A question for the mathematicians

What if the two missing squares have *different* colors, like on this board? Can it be covered? Do you think that is *always* true, for any two squares of different colors?

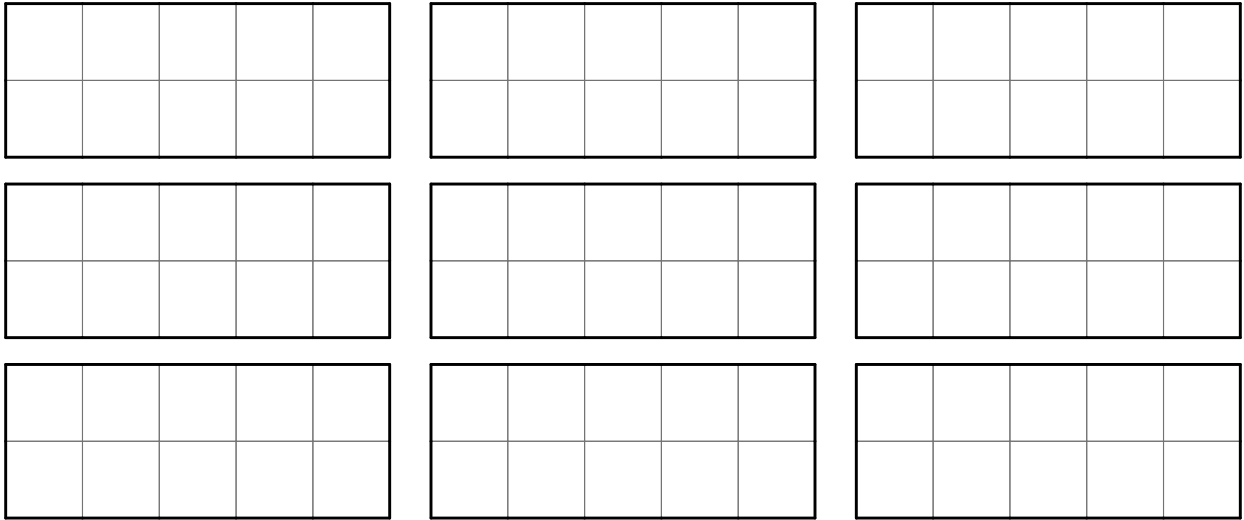


7. More ways

Before you draw anything, guess: how many different ways are there to cover a 2×4 board? _____
Now draw every different way.



Now a 2×5 board.

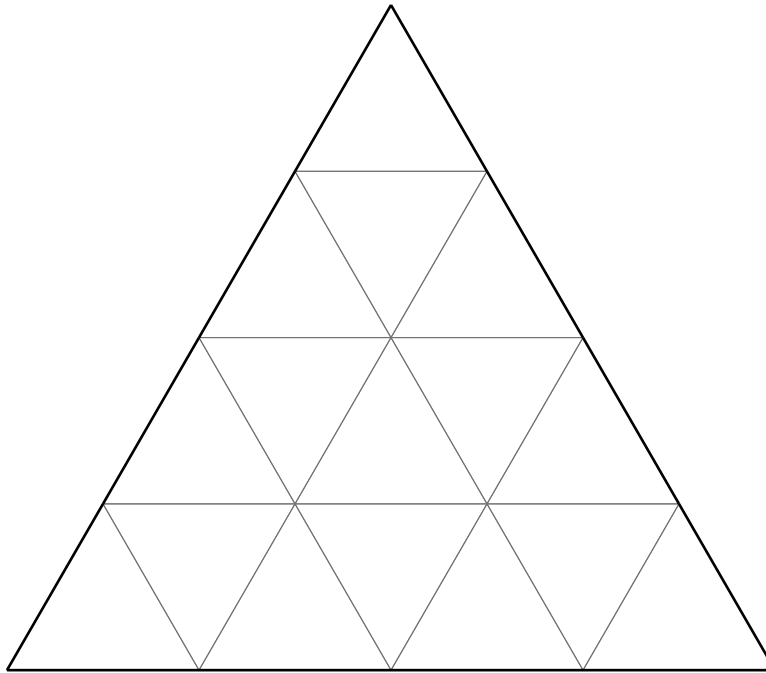


board	2×1	2×2	2×3	2×4	2×5	2×6
number of ways	1					

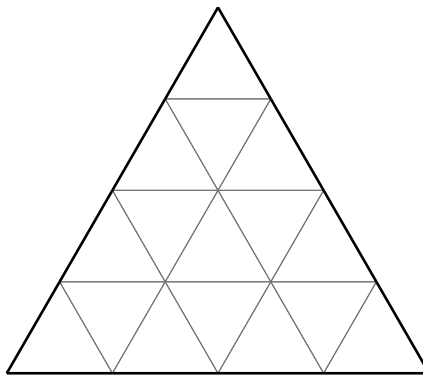
Do you see a pattern? Use it to predict the 2×6 board. Then explain why the pattern happens. Hint: look at the left end of the board. What are the only two things that can happen there?

8. Back to the blocks

This triangle holds 16 little triangles, an even number. Can you fill it with blue rhombi only? Try it with the pattern blocks. (The outline is at true size.)



Now shade every little triangle that points *up*, and count.



pointing up: _____ pointing down: _____

One blue rhombus always covers: _____

Is there *any* triangle of this kind that can be filled with blue rhombi only? Why?

Paper dominoes

Cut along the heavy lines.

